Architectural Blueprint and Component Analysis for an AI-Powered Blast Radius Analyzer

The engineering of an autonomous, AI-powered Blast Radius Analyzer represents a highly complex intersection of disparate computer science domains. To successfully crawl live web applications, parse unstructured product requirements, construct an Evidence-backed Product Knowledge Graph, and perform graph-based impact analysis on pull requests, a system must seamlessly orchestrate web automation, document parsing, static code analysis, and advanced graph retrieval architectures.

An exhaustive technical analysis of the open-source ecosystem reveals that constructing these subsystems from scratch is entirely unnecessary. The open-source community provides production-grade, highly active repositories that solve the foundational challenges of Abstract Syntax Tree (AST) querying, hierarchical Knowledge Graph generation, probabilistic entity resolution, and agent-computer interfacing. The following report systematically deconstructs these repositories to formulate a comprehensive implementation blueprint for the Blast Radius Analyzer.

Top 50 Repositories Ranked by Relevance

The following structured assessment ranks the fifty most relevant open-source projects for the specified architecture. The ranking prioritizes repositories that directly solve sub-problems within crawler automation, parsing, graph construction, static analysis, and agentic pull request evaluation[1].

Rank	Repository Name	Primary Domain	Core Utility for Blast Radius Analyzer
1	microsoft/graphrag	Knowledge Graph	Hierarchical community detection and graph retrieval pipeline.
2	joernio/joern	Code Analysis	Construction of Code Property Graphs (CPGs) for impact tracing.

3	docling-project/docling	Document Parsing	Translating PRDs, wikis, and PDFs into structured knowledge.
4	ast-grep/ast-grep	Static Analysis	Tree-sitter based structural code search for localized impact.
5	unclecode/crawl4ai	Web Crawling	LLM-friendly web crawling producing RAG-ready Markdown.
6	HKUDS/LightRAG	Graph Retrieval	Dual-layer (vector and graph) retrieval for dynamic architectures.
7	browser-use/browser-use	UI Understanding	Playwright-based autonomous agents for UI and DOM analysis.
8	qodo-ai/pr-agent	PR Analysis	LLM-based automated pull request review and explainability.
9	All-Hands-AI/OpenHands	AI Agents	Autonomous agent-computer interface (ACI) for repository tasks.
10	moj-analytical-services/splink	Entity Resolution	Probabilistic linkage bridging product requirements to source code.

11	skyvern-ai/skyvern	UI Understanding	Computer vision-based autonomous web workflow automation.
12	princeton-nlp/SWE-agent	AI Agents	Automated GitHub issue resolution and software engineering tasks.
13	lavague-ai/LaVague	Web Agents	Web agent framework parsing DOMs into actionable tasks.
14	crytic/slither	Static Analysis	Reference implementation for AST-based vulnerability traversal.
15	langchain-ai/langchain	LLM Framework	Foundational orchestration for RAG and agentic workflows.
16	run-llama/llama_index	Knowledge Integration	Data framework bridging structured and unstructured software data.
17	BerriAI/litellm	AI Gateway	Unified API proxy for orchestrating multiple LLM providers.
18	Aider-AI/aider	AI Agents	Git-native terminal pair programming agent architecture.

19	langgenius/dify	App Framework	Visual workflow orchestration for connecting LLM nodes.
20	openai/codex	AI Agents	Containerized, sandboxed execution environment for code analysis.
21	logspace-ai/langflow	Pipeline Builder	UI for visualizing GraphRAG and multi-agent pipelines.
22	infiniflow/ragflow	RAG Engine	Deep document understanding with complex chunking strategies.
23	lobehub/lobe-chat	Explainable UI	Chat interface framework for surfacing blast radius reports.
24	embedchain/embedchain	RAG Framework	Memory layer abstraction for application context.
25	deepset-ai/haystack	RAG Framework	Orchestration for advanced retrieval augmented generation.
26	mindsdb/mindsdb	AI Database	Integrating AI models directly with underlying data sources.

27	Cinnamon/kotaemon	RAG Chat	Open-source RAG UI for interacting with documentation.
28	songquanpeng/one-api	LLM Gateway	Key management and load balancing for high-volume LLM calls.
29	labring/FastGPT	KG App Platform	Knowledge-based platform for deploying question-answering systems.
30	nomic-ai/gpt4all	Local LLMs	Local execution of models for sensitive repository data.
31	comfyanonymous/ComfyUI	Pipeline Builder	Node-based workflow execution (reference for visual pipelines).
32	chatchat-space/Langchain-Chatchat	Local RAG	Offline RAG pipeline for highly secure codebase analysis.
33	go-skynet/LocalAI	Inference	OpenAI-compatible API for localized model deployment.
34	flairNLP/flair	NLP Framework	Base entity extraction for parsing software requirements.

35	xenova/transformers.js	Web Inference	Running LLM inference inside browser-based report viewers.
36	netease-youdao/QAnything	Enterprise RAG	Cross-modal Q&A architecture for mixed software assets.
37	h2oai/h2ogpt	Private LLM	Privacy-first chat interface for proprietary codebases.
38	pathwaycom/llm-app	Data Pipelines	Real-time syncing with dynamic GitHub and Jira data sources.
39	stitionai/devika	AI Agents	Autonomous software developer architecture.
40	sola-st/repairagent	AI Agents	Autonomous bug repair agent evaluation framework.
41	Forethought-Technologies/AutoChain	Agent Orchestration	Lightweight, extensible framework for testing AI agents.
42	LAION-AI/Open-Assistant	Conversational AI	Chat-based interactions with third-party software systems.

43	Mintplex-Labs/anything-llm	Knowledge App	Desktop and Docker app with built-in RAG and agent builders.
44	ParisNeo/lollms-webui	Multi-modal UI	Interface for interacting with varying code and image inputs.
45	OpenBMB/UltraRAG	RAG Framework	Low-code framework building complex RAG pipelines.
46	Agenta-AI/agenta	LLMOps	Prompt management and evaluation for the PR analysis agents.
47	0hq/WebGPT	Client Inference	WebGPU-based inference for offline code reasoning.
48	23blocks-OS/ai-maestro	Agent Skills	Orchestrator providing code graph queries to sub-agents.
49	2dogsandanerd/ClawRag	Vector Integration	Combines Docling document processing with ChromaDB.
50	BoundaryML/baml	Prompt Engineering	Strongly typed framework bridging LLM output to backend logic.

Exhaustive Repository Technical Analysis

The following subsections provide a rigorous deconstruction of the most critical repositories. Each analysis details the metadata, technical stack, architectural patterns, and the precise methodology required to integrate the repository into the overarching Blast Radius Analyzer architecture.

Crawl4AI

Metadata Parameter	Details
Repository Name	unclecode/crawl4ai
GitHub URL	github.com/unclecode/crawl4ai
Stars	~73.8k[1]
Forks	~7.6k[1]
License	Apache-2.0[1]
Last Updated	Active (2026)[12]
Primary Language	Python, HTML[14]
Maintainer	unclecode[1]
Installation Complexity	Low (pip install crawl4ai or Docker)[1]
Project Status	Highly Active[1]

Technical Parameter	Details
Overall Purpose	LLM-friendly web crawler extracting structural data into Markdown.

System Architecture	Asynchronous execution pipeline wrapping headless browsers.
Folder Structure	Python modules containing strategies for chunking and extraction.
Important Modules	AsyncWebCrawler, JsonXPathExtractionStrategy, PruningContentFilter.
Technologies Used	Python asyncio, Playwright.
Database	Caching layer enabled via CrawlerRunConfig[11].
Parsing Libraries	Heuristic HTML parsers, XPath, CSS selectors.
Graph Technologies	Generates node-ready semantic clusters via hierarchical chunking.
LLM Framework	LLM Extraction Strategies via OpenAI-compatible endpoints[13].
Agent Framework	None natively, functions as a tool for external agents[13].

The architectural patterns present in Crawl4AI provide a foundational reference for the autonomous website crawler subsystem. The repository achieves high performance by utilizing an asynchronous architecture built on Python's asyncio and Playwright, enabling the concurrent crawling of modern Single Page Applications (SPAs)[1]. The implemented algorithms prioritize noise reduction, utilizing heuristic-based filtering mechanisms such as the PruningContentFilter to strip irrelevant DOM elements before they consume valuable LLM context windows[11]. Furthermore, the repository implements cosine clustering for semantic document filtering and topic-based chunking, ensuring the output is perfectly formatted for Retrieval-Augmented Generation (RAG) pipelines[14].

The exact components that can be directly reused include the AsyncWebCrawler core and the LLM-friendly Markdown extraction pipelines, which are essential for traversing live web applications to map user flows[1]. When building the data ingestion layer, the LLMExtractionStrategy and JsonXPathExtractionStrategy files should be studied first, as they demonstrate how to pass specific instructions to an LLM to refine the extraction of UI

elements[11]. A notable limitation of the repository is its heavy reliance on underlying browser binaries via Playwright, which can introduce instability in strict, air-gapped Docker environments or Kubernetes clusters requiring extensive proxy configurations[1]. Despite this limitation, the repository is highly recommended as a direct dependency for the initial stage of constructing the Evidence-backed Product Knowledge Graph.

Browser-use

Metadata Parameter	Details
Repository Name	browser-use/browser-use
GitHub URL	github.com/browser-use/browser-use
Stars	105.2k[16]
Forks	11.5k[17]
License	MIT[16]
Last Updated	Active (2026)[16]
Primary Language	Python (99.2%)[18]
Maintainer	browser-use organization[19]
Installation Complexity	Low (uv pip install browser-use)[17]
Project Status	Highly Active[19]

Technical Parameter	Details
Overall Purpose	Enables AI agents to autonomously control web browsers.
System Architecture	Decoupled agent logic and browser session contexts.

Folder Structure	/skills, /open-source, core Python library modules[17].
Important Modules	BrowserSession, Agent, ChatBrowserUse.
Technologies Used	Python >= 3.11, asyncio, Playwright[17].
Database	Persistent filesystem memory for session management[18].
Parsing Libraries	Custom accessibility tree extraction.
Graph Technologies	None natively.
LLM Framework	Provider-agnostic via ChatBrowserUse (Anthropic, OpenAI, Google)[18].
Agent Framework	Internal Agent class orchestrating tasks via a system prompt[18].

The browser-use repository solves the critical problem of dynamically mapping complex, authenticated user interfaces that static crawlers cannot penetrate. The architectural pattern separates the BrowserSession context from the autonomous Agent logic, allowing multiple LLM-driven agents to operate concurrently on isolated browser instances[17]. The core algorithm relies on extracting an accessibility tree from the live DOM, compressing the visual layout into a token-efficient semantic representation that allows the LLM to deduce where to click, type, or navigate[18]. This design idea—translating a visual DOM into a structured semantic tree—is a pattern worth borrowing for the UI Understanding phase of the Blast Radius Analyzer.
Direct reuse of the Python library is highly recommended for mapping interactive user flows and extracting stateful UI components. The source files defining the Agent and ChatBrowserUse abstractions should be studied first, as they reveal how to handle provider-prefixed model IDs and inject custom system messages without overriding the default browser control instructions[17]. The primary limitation of this approach is inference latency; because every action requires a sequential LLM call to evaluate the accessibility tree, it is significantly slower than static DOM parsing[18]. Consequently, browser-use is recommended as a direct dependency specifically for the deep UI Analysis and dynamic flow extraction subsystems, supplementing the faster static crawling mechanisms.

Docling

Metadata Parameter	Details
Repository Name	docling-project/docling
GitHub URL	github.com/docling-project/docling
Stars	63.5k[20]
Forks	4.5k[20]
License	MIT[20]
Last Updated	Active (2026)[5]
Primary Language	Python[5]
Maintainer	IBM Research (docling-project)[21]
Installation Complexity	Low (pip install docling)[5]
Project Status	Highly Active[5]

Technical Parameter	Details
Overall Purpose	Unified document parsing for generative AI ecosystems.
System Architecture	Multi-modal processing pipeline outputting standard representations.
Folder Structure	Modular converters for PDF, HTML, Markdown, and visual media.
Important Modules	DocumentConverter, DoclingDocument, ProvenanceItem.

Technologies Used	Python 3.10+, PyTorch, OCR, Audio ASR.
Database	None natively (exports to lossless JSON/Markdown)[5].
Parsing Libraries	Custom layout analysis, table structure, and reading order engines.
Graph Technologies	Ecosystem includes docling-graph for KG generation[23].
LLM Framework	Integrates Visual Language Models (VLMs) like GraniteDocling[5].
Agent Framework	Exposes MCP server and integrates with LangChain/LlamaIndex[5].

The Docling repository provides an unparalleled architecture for ingesting the vast array of unstructured formats that comprise product documentation. The system architecture is built around a unified, expressive representation format known as the DoclingDocument, which acts as a normalized schema for data extracted from PDFs, Word documents, PowerPoint presentations, and images[5]. The most valuable algorithm implemented within Docling is its provenance tracking mechanism. The ProvenanceItem schema meticulously records the exact physical location—including page number, bounding box (bbox), and character span—of every extracted data point, tying the semantic text back to its physical origin[24]. This design idea is critical for generating explainable Blast Radius Reports, as it allows the final AI agent to explicitly cite the exact location in a Product Requirements Document (PRD) that justifies an impact warning.

The DocumentConverter module can be directly reused as the universal ingestion engine for all documentation parsing[5]. Developers should first study the data models defining the DoclingDocument and the recent architectural proposals for a generalized ProvenanceType union, which expands tracking beyond spatial layouts to temporal media like video timestamps[24]. While the core parsing engine is highly robust, deploying the advanced Visual Language Model (VLM) features or the Optical Character Recognition (OCR) engines requires significant computational resources, leading to the distribution of heavily tagged container images (e.g., CUDA 13.0 variants)[22]. Docling is strongly recommended as a direct dependency for the Requirements Traceability and Documentation Parsing subsystems.

ast-grep

Metadata Parameter	Details

Repository Name	ast-grep/ast-grep
GitHub URL	github.com/ast-grep/ast-grep
Stars	15.1k[10]
Forks	416[10]
License	MIT[10]
Last Updated	Active (2026)[10]
Primary Language	Rust (94.8%)[10]
Maintainer	ast-grep organization[10]
Installation Complexity	Low (via npm, pip, brew, or cargo)[10]
Project Status	Highly Active[10]

Technical Parameter	**Details**
Overall Purpose	High-performance structural code search and AST rewriting.
System Architecture	Multi-threaded AST traversal engine utilizing Tree-sitter.
Folder Structure	Rust crates (crates/cli), rule configuration directories (rules/).
Important Modules	YAML rule engine, Model Context Protocol (MCP) server.
Technologies Used	Rust, Tree-sitter[10].

Database	None natively.
Parsing Libraries	Tree-sitter language bindings (C, Java, Python, TypeScript, etc.)[10].
Graph Technologies	Abstract Syntax Tree (AST) navigation.
LLM Framework	None natively in the core, but integrates via the ast-grep-mcp server[26].
Agent Framework	Provides agent skills (e.g., Claude Code plugins) for autonomous search[25].

The ast-grep repository introduces a paradigm shift in static code analysis by allowing developers and AI agents to query codebases using isomorphic syntax patterns rather than abstract graph query languages or brittle regular expressions. The architecture utilizes the Tree-sitter parsing library to generate an Abstract Syntax Tree (AST), over which ast-grep executes multi-threaded structural matching algorithms[10]. The system allows for the definition of complex relational constraints—such as verifying if a matched node resides inside, precedes, or follows another node—entirely via YAML configurations[10]. This design provides an incredibly token-efficient mechanism for Large Language Models to explore repository architectures without ingesting massive raw file contexts[26].

The core binary, the YAML rule engine, and specifically the experimental Model Context Protocol (MCP) server (ast-grep-mcp) can be directly reused to form the backbone of the Static Code Analysis subsystem[10]. By equipping an AI agent with the ast-grep MCP server, the agent can dynamically write search rules, test them against the AST, and map the precise call graphs associated with a modified function in a pull request[25]. When studying the repository, engineers should focus on the YAML rule configuration schemas and the MCP server implementation logic, as these represent the primary interfaces for AI integration[26]. While ast-grep is profoundly fast, its limitation lies in cross-file data-flow tracking, as ASTs inherently represent single-file syntactic structures rather than global semantic dependencies. Nonetheless, ast-grep is recommended as a direct dependency for structural search and localized Call Graph Construction.

Joern

Metadata Parameter	Details
Repository Name	joernio/joern

GitHub URL	github.com/joernio/joern
Stars	3.3k[29]
Forks	430[29]
License	Apache-2.0[29]
Last Updated	Active (2026)[29]
Primary Language	Scala (98.1%)[29]
Maintainer	joernio (industry partners and researchers)[29]
Installation Complexity	High (requires JDK 21, sbt, C++ compilation toolchains)[29]
Project Status	Active (Migrating to Flatgraph)[29]

Technical Parameter	**Details**
Overall Purpose	Vulnerability discovery via Code Property Graph (CPG) analysis.
System Architecture	Multi-language frontends parsing to a unified graph database.
Folder Structure	Language-specific AST generators (astgen-monorepo)[31].
Important Modules	javasrc2cpg, ghidra2cpg, CPGQL interactive shell.
Technologies Used	Scala 3, JVM.
Database	Flatgraph (Custom columnar NoSQL graph storage)[29].

Parsing Libraries	Custom language parsers translating to CPG specifications.
Graph Technologies	Code Property Graphs (fusing AST, CFG, and PDG).
LLM Framework	None natively.
Agent Framework	None natively.

The Joern repository provides a masterclass in deep, semantic static analysis through the construction of Code Property Graphs (CPGs). A CPG mathematically fuses the Abstract Syntax Tree (AST), Control Flow Graph (CFG), and Program Dependence Graph (PDG) into a single, unified data structure[29]. The architectural pattern involves language-specific frontends that parse source code and emit intermediate representations, which are then loaded into Flatgraph, a highly optimized, columnar NoSQL graph database written in Scala[29]. The implemented algorithms allow for complex data-flow tracking, semantic reachability, and taint analysis, enabling queries that can track how an arithmetic expression in one function propagates to a memory allocation call elsewhere in the codebase[33].

While the theoretical depth of Joern is unmatched for analyzing software architecture, direct reuse in a lightweight, LLM-driven Blast Radius Analyzer is challenging. The reliance on the JVM, the steep learning curve of the Scala-based domain-specific query language, and the massive computational overhead required to generate CPGs present significant architectural friction[29]. Consequently, Joern is recommended strictly as a reference implementation. The design ideas defining how control flow and data flow can be mathematically represented as graph edges should be borrowed to inform the schema design of the overarching Product Knowledge Graph.

Microsoft GraphRAG

Metadata Parameter	Details
Repository Name	microsoft/graphrag
GitHub URL	github.com/microsoft/graphrag
Stars	34.7k[3]
Forks	3.7k[3]

License	MIT[3]
Last Updated	Active (2026)[3]
Primary Language	Python (88.4%)[3]
Maintainer	Microsoft Research[3]
Installation Complexity	Moderate (pip install graphrag + environment/API configuration)[3]
Project Status	Highly Active[3]

Technical Parameter	**Details**
Overall Purpose	Enhancing LLM reasoning via knowledge graph memory structures.
System Architecture	Multi-stage indexing pipeline generating hierarchical graphs.
Folder Structure	Workspaces containing .env and settings.yaml for pipeline configs[35].
Important Modules	Indexing pipeline, Global Search, Local Search.
Technologies Used	Python, OpenAI/Azure API integrations.
Database	Agnostic (can emit to standard graph or relational formats).
Parsing Libraries	LLM-driven entity and relationship extraction.
Graph Technologies	NetworkX, hierarchical graph clustering.

LLM Framework	OpenAI, Azure OpenAI[35].
Agent Framework	Orchestrates internal prompting pipelines for data extraction.

The GraphRAG architecture pioneered by Microsoft Research represents a fundamental evolution in how large language models interact with private datasets. Instead of relying purely on vector similarity—which fails when answering global questions that require synthesizing information across multiple documents—the system uses the LLM to extract a dense network of entities and relationships[3]. The most crucial algorithm implemented is the Leiden community detection algorithm, which partitions the massive knowledge graph into hierarchical communities[36]. By generating textual summaries for each community, the system can perform a "Global Search" that reasons across the entire dataset simultaneously[35].

The graph extraction pipelines and the distinct query routing mechanisms (specifically the bifurcation of global and local search strategies) can be directly reused for constructing the Evidence-backed Product Knowledge Graph[35]. Engineers should study the prompt tuning guides and the pipeline configuration mechanics, as tuning the LLM extraction prompts is necessary for accurately identifying software-specific entities like "API Endpoints" or "UI Components"[3]. The primary limitation of Microsoft's GraphRAG is the extreme computational cost and time required for the indexing phase, as every document chunk must be processed by the LLM to extract relationships[3]. Because of this, it is recommended as a direct dependency for the initial baseline knowledge graph construction, but may require supplementation for real-time incremental updates.

LightRAG

Metadata Parameter	Details
Repository Name	HKUDS/LightRAG
GitHub URL	github.com/HKUDS/LightRAG
Stars	5.3k[39]
Forks	198[39]
License	MIT[4]
Last Updated	Active (2026)[39]

Primary Language	Python[4]
Maintainer	HKUDS (Zirui Guo)[4]
Installation Complexity	Low (pip install lightrag-hku)[37]
Project Status	Highly Active[40]

Technical Parameter	**Details**
Overall Purpose	Fast, scalable, and incremental knowledge graph RAG framework.
System Architecture	Dual-layer architecture managing vector embeddings and graph structures.
Folder Structure	Abstracted storage backends (llm/llama_index_impl.py, etc.)[40].
Important Modules	Query routers, incremental graph merger, storage abstractions.
Technologies Used	Python 3.10+, LlamaIndex integration[4].
Database	MongoDB, PostgreSQL (pgvector/Apache AGE), Neo4j, Oracle 23ai[40].
Parsing Libraries	Document chunking and entity extraction via LLMs.
Graph Technologies	Set merging algorithms, directional multi-hop traversals[4].
LLM Framework	OpenAI, Ollama, Hugging Face, LiteLLM[40].
Agent Framework	None natively, functions as a retrieval backend.

LightRAG directly addresses the computational bottlenecks inherent in Microsoft's GraphRAG

by implementing an architecture optimized for rapid adaptation to dynamic data. The system utilizes a dual-layer architecture that maintains both a vector database for semantic similarity and a graph database for complex semantic dependencies[4]. The most innovative algorithmic approach is its incremental update mechanism. When new documentation or code is analyzed, LightRAG generates a localized graph of the new data and utilizes set merging algorithms to integrate it directly into the global graph, bypassing the need for a total index rebuild[4]. This pattern is essential for a Blast Radius Analyzer, which must rapidly ingest the delta of a pull request without pausing for hours of re-indexing.

The abstracted storage layer and the five-mode query routing system (local, global, hybrid, naive, mix) represent components that can be directly reused[4]. Specifically, the backend implementations that allow the system to utilize PostgreSQL as a one-stop solution for Key-Value storage, vector storage (via pgvector), and graph storage (via Apache AGE) provide immense architectural simplicity for enterprise deployments[40]. Engineers should study the implementation of the mix retrieval mode, which merges traditional text chunk retrieval with global graph relationship contexts to yield the most comprehensive answers[4]. LightRAG is highly recommended as a direct dependency for the Graph Construction Pipeline and Graph-based Retrieval subsystems, serving as a highly optimized alternative to Microsoft GraphRAG.

Splink

Metadata Parameter	Details
Repository Name	moj-analytical-services/splink
GitHub URL	github.com/moj-analytical-services/splink
Stars	Not explicitly enumerated in source, widely adopted[43]
Forks	Not explicitly enumerated[43]
License	MIT[43]
Last Updated	Active (2026)[43]
Primary Language	Python[43]
Maintainer	UK Ministry of Justice (moj-analytical-services)[43]

Installation Complexity	Low (pip install splink[duckdb])[43]
Project Status	Highly Active[43]

Technical Parameter	**Details**
Overall Purpose	Scalable probabilistic data linkage and entity resolution.
System Architecture	Statistical modeling engine executing over distributed SQL backends.
Folder Structure	Backend-specific implementation modules (DuckDB, Spark, Postgres).
Important Modules	Fellegi-Sunter model implementation, interactive visualization suite.
Technologies Used	Python 3.9+, SQL.
Database	DuckDB, Apache Spark, PostgreSQL[43].
Parsing Libraries	None natively (operates on pre-parsed tabular data).
Graph Technologies	Outputs linked clusters that can form graph edges[43].
LLM Framework	None natively.
Agent Framework	None natively.

The Splink repository solves a profoundly difficult problem within the Knowledge Fusion subsystem: probabilistic entity resolution. When building a product knowledge graph, an architectural component might be referred to as the "User Authentication Module" in a PRD, but as auth_controller.ts in the source code. Splink provides the statistical engine to link these disparate records without requiring a unique identifier[43]. The architecture implements the Fellegi-Sunter mathematical model for record linkage, utilizing unsupervised learning

algorithms (specifically Expectation-Maximization) to train the linkage models without requiring human-labeled training datasets[43].
The probabilistic matching engine can be directly reused to construct the edges between the software requirements nodes and the source code nodes within the overarching graph. The ability to execute this logic natively in Python using in-memory databases like DuckDB, or scaling up to Apache Spark for massive repositories, provides excellent architectural flexibility[43]. The primary limitation is that Splink operates optimally on datasets with multiple, non-correlated columns (e.g., name, date, location), rather than single columns containing a "bag of words"[43]. Therefore, the data extracted by Docling and ast-grep must be properly structured into distinct metadata columns before being passed to Splink. It is recommended as a direct dependency for the Entity Resolution subsystem.

PR-Agent (Qodo Merge)

Metadata Parameter	Details
Repository Name	qodo-ai/pr-agent
GitHub URL	github.com/qodo-ai/pr-agent
Stars	12.2k[2]
Forks	1.6k[2]
License	Polyform/Open-source dual licensing models[47]
Last Updated	Active (2026)[2]
Primary Language	Python (99.8%)[47]
Maintainer	Qodo (formerly CodiumAI)[47]
Installation Complexity	Moderate (Requires Webhook/GitHub App configuration)[46]
Project Status	Highly Active[2]

Technical Parameter	Details
Overall Purpose	Automated pull request analysis, feedback, and suggestions.
System Architecture	Event-driven webhook listeners invoking targeted LLM analysis tools.
Folder Structure	CLI tools (pr_agent.cli), configuration files (.pr_agent.toml)[46].
Important Modules	/review, /describe, /improve tools, LiteLLM integration.
Technologies Used	Python, LiteLLM, Git provider APIs.
Database	None strictly required for open-source version (operates statelessly).
Parsing Libraries	Git diff parsers and abstract syntax filters.
Graph Technologies	None natively.
LLM Framework	LiteLLM supporting OpenAI, Gemini, Claude, and local models[46].
Agent Framework	Orchestrates single-call LLM interactions based on JSON prompts[47].

The PR-Agent repository establishes a robust architectural pattern for integrating generative AI directly into the software development lifecycle. The system is designed to respond to Git provider webhooks (GitHub, GitLab, Bitbucket), invoking highly specialized tools to analyze incoming pull requests[46]. The most critical algorithmic design idea within PR-Agent is its "PR Compression strategy." Large pull requests generate massive git diff outputs that easily overwhelm LLM context windows; PR-Agent implements algorithms to intelligently truncate, summarize, and distill these diffs into a dense representation that preserves the semantic intent of the changes[47]. Furthermore, the system enforces a strict JSON prompting strategy, ensuring the LLM outputs modular, customizable data that can be programmatically parsed and posted back as a GitHub Check or comment[2].

The webhook listener architecture and the PR Compression algorithms should be studied extensively as reference implementations. While the open-source version provides excellent

baseline code review, injecting complex, multi-step graph queries to determine a "Blast Radius" requires significant modification of the core tool logic, as the repository is optimized for single-call LLM interactions rather than deep, iterative agentic reasoning[47]. Furthermore, some advanced features, like persistent suggestion tracking and incremental updates for subsequent commits, are locked behind the proprietary Qodo Merge tier[51]. Therefore, PR-Agent is recommended primarily as an inspiration and reference implementation for handling the Pull Request Ingestion phase.

OpenHands

Metadata Parameter	Details
Repository Name	All-Hands-AI/OpenHands
GitHub URL	github.com/All-Hands-AI/OpenHands
Stars	~75k[8]
Forks	High (Highly active community)[52]
License	MIT[52]
Last Updated	Active (2026)[52]
Primary Language	Python[53]
Maintainer	All-Hands-AI[52]
Installation Complexity	Moderate (Requires Docker for sandboxed execution)[52]
Project Status	Highly Active[52]

Technical Parameter	Details
Overall Purpose	Autonomous AI software developer platform and agent-computer interface.

System Architecture	Parallelized, multi-agent delegation within sandboxed containers.
Folder Structure	Agent SDKs (openhands/tools), CLI modules, local GUI[52].
Important Modules	Agent-Computer Interface (ACI), File Editor Tool, Shell Executor[54].
Technologies Used	Python, Docker, React (for GUI)[52].
Database	None strictly required for stateless agent runs.
Parsing Libraries	Tree-sitter for built-in code linting[54].
Graph Technologies	None natively.
LLM Framework	Provider-agnostic support for 100+ LLMs (Claude, GPT, Ollama)[52].
Agent Framework	Custom autonomous reasoning and task decomposition engine[52].

The OpenHands repository provides the foundational reasoning engine required to synthesize the outputs of the static analysis tools and the knowledge graph into a cohesive Blast Radius Report. The architecture revolves around an Agent-Computer Interface (ACI), which grants the LLM secure, sandboxed access to execute terminal commands, edit files, and browse the web within a controlled Docker container[52]. The implemented algorithms handle complex task decomposition, allowing the primary agent to delegate specific coding or analysis tasks to specialized sub-agents (e.g., spawning a sub-agent to navigate a specific directory structure)[52]. Crucially, the system implements advanced context management techniques—such as signal-driven context injection and insertion-time tool result trimming—to prevent the agent from entering infinite loops or overflowing its context window during deep repository traversals[52].

The openhands-aci and the core Agent SDK can be directly reused to act as the central orchestration brain of the Blast Radius Analyzer[53]. By deploying the OpenHands agent into a sandbox containing the cloned repository and equipping it with the ast-grep MCP server, the agent can autonomously investigate the ripple effects of a code change in real-time[26]. Engineers should study the implementation of the tool-call loop guards and the multi-agent delegation patterns[52]. The primary limitation is the inherent latency and overhead of spinning

up and managing Docker containers for every pull request analysis, which requires robust underlying infrastructure[8]. OpenHands is highly recommended as a direct dependency for the AI Agents for Software Engineering subsystem.

Skyvern

Metadata Parameter	Details
Repository Name	skyvern-ai/skyvern
GitHub URL	github.com/skyvern-ai/skyvern
Stars	~22k[55]
Forks	~2k[56]
License	AGPL-3.0[55]
Last Updated	Active (2026)[55]
Primary Language	Python[55]
Maintainer	Skyvern AI[56]
Installation Complexity	Moderate (Requires PostgreSQL, Rust, Playwright configurations)[55]
Project Status	Highly Active[55]

Technical Parameter	Details
Overall Purpose	Computer vision-based autonomous web workflow automation.
System Architecture	Swarm of agents utilizing visual reasoning and Playwright SDKs.

Folder Structure	Workflows, blocks, and API endpoints[57].
Important Modules	Skyvern client, Session manager, Workflow Builder[55].
Technologies Used	Python 3.11+, PostgreSQL, Playwright, Computer Vision.
Database	PostgreSQL[55].
Parsing Libraries	Visual layout analysis bypassing traditional DOM parsing[55].
Graph Technologies	None natively.
LLM Framework	Integrates Vision-Language Models (VLMs)[55].
Agent Framework	Task-Driven autonomous agent design (inspired by BabyAGI/AutoGPT)[55].

Skyvern introduces a radically different architectural approach to web automation by eschewing traditional DOM parsing and XPath selectors in favor of Vision-Language Models (VLMs) and computer vision algorithms. The system utilizes a swarm of agents that "look" at the rendered webpage to plan and execute interactions, making it highly resistant to underlying HTML layout changes that typically break standard automation scripts[55]. This architectural pattern—visually reasoning about the state of a web application—is theoretically ideal for mapping the User Experience layers of the Product Knowledge Graph, bridging the gap between how a UI component is described in a PRD and how it actually renders in production. However, despite its innovative approach, the repository possesses limitations that impact its viability as a direct dependency for this specific assignment. The continuous processing of high-resolution images by VLMs introduces significant latency and extreme API costs[57]. Furthermore, the AGPL-3.0 license mandates stringent copyleft requirements that can complicate deployment in closed-source enterprise environments[55]. Therefore, Skyvern is recommended strictly as an inspiration. The concepts underlying its visual reasoning engine should be studied to understand how future iterations of the crawler subsystem might evolve beyond pure HTML parsing.

SWE-agent

Metadata Parameter	Details

Repository Name	princeton-nlp/SWE-agent
GitHub URL	github.com/princeton-nlp/SWE-agent
Stars	~15k[9]
Forks	Not explicitly enumerated in source[9]
License	MIT (Implied via standard academic OS releases, though check required)
Last Updated	Active (2026)[9]
Primary Language	Python[9]
Maintainer	Princeton NLP[9]
Installation Complexity	Moderate (Requires sandboxed environments for execution)
Project Status	Active (Academic/Research tool)[9]

Technical Parameter	**Details**
Overall Purpose	Autonomous resolution of GitHub issues via an agentic terminal.
System Architecture	Tightly optimized feedback loop between LLM and a sandboxed shell.
Folder Structure	Academic research structure with evaluation harnesses.
Important Modules	Terminal interaction environment, SWE-bench evaluator.
Technologies Used	Python, Docker.

Database	None natively.
Parsing Libraries	Custom terminal output formatters[59].
Graph Technologies	None natively.
LLM Framework	Agnostic (Tested heavily with GPT-4)[59].
Agent Framework	Custom agent loop optimized for the SWE-bench evaluation[59].

The SWE-agent repository, developed by Princeton NLP, serves as the definitive academic benchmark for evaluating AI agents on software engineering tasks. The architecture is defined by an incredibly optimized feedback loop that manages how the LLM interacts with a simulated terminal[9]. Rather than simply dumping raw command-line output back to the LLM, the system implements specific design ideas to paginate, format, and truncate outputs (such as directory listings or file reads), ensuring the model maintains deep contextual awareness without hallucinating due to information overload[59].

While OpenHands provides a more production-ready, feature-rich platform for the final deployment, the architectural patterns within SWE-agent should be studied extensively. Specifically, engineers should examine the code files responsible for formatting the terminal feedback loop, as these techniques can be directly applied to optimize how the results of an ast-grep graph query are presented to the reporting agent during the Blast Radius analysis phase[58]. SWE-agent is recommended as a highly valuable reference implementation.

Slither

Metadata Parameter	Details
Repository Name	crytic/slither
GitHub URL	github.com/crytic/slither
Stars	6.3k[60]
Forks	1.1k[60]
License	AGPL-3.0[60]

Last Updated	Active (2026)[60]
Primary Language	Python (>=3.10)[60]
Maintainer	Trail of Bits (crytic)[60]
Installation Complexity	Low (uv tool install slither-analyzer)[60]
Project Status	Highly Active[60]

Technical Parameter	**Details**
Overall Purpose	Static analysis and vulnerability detection for smart contracts.
System Architecture	Multi-pass static analysis engine leveraging intermediate representations.
Folder Structure	Python modules containing distinct vulnerability detectors (detectors/)[60].
Important Modules	SlithIR generation, Custom Printer API[60].
Technologies Used	Python, solc compiler integration[60].
Database	None natively.
Parsing Libraries	Relies on underlying compiler AST outputs.
Graph Technologies	Intermediate representation graphs (SlithIR)[60].
LLM Framework	None natively.
Agent Framework	None natively.

Although Slither is domain-specific (targeting Solidity and Vyper), its system architecture provides a masterclass in the design of static analysis frameworks. The core algorithm relies on

translating the raw Abstract Syntax Tree (AST) produced by the compiler into a proprietary intermediate representation known as SlithIR[60]. This normalization process simplifies the underlying logic, allowing the system to execute complex data-flow tracking, variable shadowing detection, and taint analysis algorithms in less than a second per file[60].
The design ideas embedded in the construction of the SlithIR should be carefully studied when architecting the Call Graph Construction subsystem. Understanding how Slither normalizes diverse syntax patterns into a standardized graph makes the detection of specific impact patterns (or vulnerabilities) trivial to implement as discrete detection modules[60]. While it cannot be used directly for a general-purpose software project (due to its language constraints and AGPL license), it stands as a premier reference implementation for structuring the internal static analysis logic of the Blast Radius Analyzer.

LaVague

Metadata Parameter	Details
Repository Name	lavague-ai/LaVague
GitHub URL	github.com/lavague-ai/LaVague
Stars	Not explicitly enumerated (Growing)[63]
Forks	Not explicitly enumerated (10 active PRs)[63]
License	Apache-2.0[63]
Last Updated	Active (2026)[63]
Primary Language	Python (70.2%), TypeScript (15.1%)[63]
Maintainer	lavague-ai[63]
Installation Complexity	Low (Python package installation)[63]
Project Status	Active[63]

Technical Parameter	Details

Overall Purpose	Web agent framework automating processes for end users.
System Architecture	Decoupled World Model and Action Engine compilation architecture.
Folder Structure	Contains testing runners, logging tools, and Gradio interfaces[63].
Important Modules	World Model, Action Engine, LaVague QA[63].
Technologies Used	Python, Selenium, Playwright, Gradio[63].
Database	None natively.
Parsing Libraries	DOM extraction into World Model state representations.
Graph Technologies	None natively.
LLM Framework	Agnostic (Contexts and configurations defined per run)[63].
Agent Framework	Custom objective-driven agent mapping states to instructions[63].

The LaVague repository provides a highly structured framework designed specifically for developers building Web Agents. The system architecture introduces a strict dichotomy between a "World Model" and an "Action Engine"[63]. The implemented algorithms allow the World Model to ingest a natural language objective alongside the current DOM state to output a set of instructions, which the Action Engine subsequently compiles into executable Selenium or Playwright code[63]. A particularly valuable architectural pattern is the LaVague QA tooling, which demonstrates how to turn BDD (Behavior-Driven Development) or Gherkin specifications directly into automated, executable UI tests[63].

This design idea—translating human-readable product requirements directly into actionable DOM navigation flows—provides a clear conceptual pathway for fusing the Product Requirements layer of the Knowledge Graph with the User Experience layer. While browser-use provides a more mature generic agent interface, LaVague's specific focus on compiling objectives into deterministic test code makes it a valuable source of inspiration for bridging the gap between PRDs and live web applications.

Component-to-Repository Mapping

The Blast Radius Analyzer architecture demands discrete but tightly integrated modules. The following mapping isolates which open-source repositories fulfill specific system requirements, identifying the primary selections and viable alternatives.

Architectural Component	Primary Repository	Alternative / Supplementary Repositories
Autonomous Web Crawling	unclecode/crawl4ai	skyvern-ai/skyvern, lavague-ai/LaVague
UI & DOM Analysis	browser-use/browser-use	skyvern-ai/skyvern (Vision-based)
Documentation Parsing	docling-project/docling	flairNLP/flair (Entity extraction)
Static Code Analysis	ast-grep/ast-grep	joernio/joern (CPGs), crytic/slither (Reference)
Knowledge Graph Construction	microsoft/graphrag	HKUDS/LightRAG (Incremental updates)
Graph Retrieval	HKUDS/LightRAG	run-llama/llama_index, langchain-ai/langchain
Entity Resolution & Fusion	moj-analytical-services/splink	Probabilistic linkage algorithms via Python
PR Analysis Ingestion	qodo-ai/pr-agent	BerriAI/litellm (Gateway routing)
Agentic Reasoning	All-Hands-AI/OpenHands	princeton-nlp/SWE-agent, Aider-AI/aider
Explainable UI / Reporting	lobehub/lobe-chat	Cinnamon/kotaemon, langgenius/dify

Recommended Repositories per Subsystem

Based on the technical breakdown, the optimal architecture relies on tools that balance analytical accuracy with ease of integration—specifically prioritizing Python and Rust ecosystems over heavy JVM dependencies.

For the **Crawler and UI Understanding** subsystem, a hybrid approach using Crawl4AI and browser-use is recommended. Crawl4AI handles the bulk of static site crawling with unparalleled speed, outputting noise-free Markdown optimized directly for the graph pipeline[1]. When the system encounters dynamic single-page applications or complex authentication walls that static scraping cannot penetrate, control is handed to the browser-use agent to map the stateful user flows[17].

For **Requirements Parsing**, Docling is the definitive choice. Product documentation rarely exists in a single, clean format. Docling homogenizes everything from text-heavy PDFs to complex architectural diagrams (via embedded VLM OCR) and outputs a standardized schema while maintaining absolute physical provenance—a critical requirement for generating explainable evidence[5].

For **Code Analysis and Call Graphs**, ast-grep is recommended. While Joern produces incredibly detailed Code Property Graphs, its Scala ecosystem and slow processing times present significant bottlenecks for rapid PR analysis[29]. Conversely, ast-grep provides instantaneous, multi-threaded structural querying utilizing Tree-sitter[10]. Crucially, its provided MCP server allows the overarching AI agent to directly query the codebase in real-time[26].

For the **Knowledge Graph Engine**, LightRAG provides the most pragmatic architecture. While Microsoft's GraphRAG yields exceptional global summaries via Leiden community detection, it requires extensive compute to re-index data[3]. A Blast Radius Analyzer requires real-time graph updates as new PRs are opened. LightRAG solves this by providing seamless incremental graph updates and supports standard backend deployments like PostgreSQL (with pgvector and Apache AGE)[4].

For **Entity Resolution**, Splink is the optimal statistical engine. Mapping a JIRA ticket mentioning "Cart Checkout" to the cart_controller.py file requires advanced fuzzy matching. Splink uses unsupervised probabilistic algorithms (Fellegi-Sunter) to scale this matching across disparate data nodes with immense accuracy[43].

Finally, for the **PR Analysis and Blast Radius Agent**, a synthesis of pr-agent and OpenHands is recommended. Qodo's pr-agent provides the structural chassis (GitHub hooks, diff compression) to ingest the code changes[47]. Instead of relying on a simple, single-pass LLM call, the compressed PR diff is handed to the OpenHands agent environment. Operating securely in a sandboxed container, the agent iteratively queries the ast-grep MCP server and the LightRAG database to mathematically deduce the impact report before pushing the final analysis back to the PR[26].

70–80% Effort Reduction Stack

Attempting to engineer AST parsers, browser automation layers, and graph extraction models from scratch requires years of dedicated engineering effort. Integrating the following three core pillars will immediately bypass roughly 80% of the implementation overhead, shifting the engineering focus from *building infrastructure* to *tuning the reasoning logic*.

1. **LightRAG paired with Docling:** This combination solves the entire data-ingestion-to-Knowledge-Graph pipeline. Docling handles the chaotic reality of unstructured enterprise data, outputting clean JSON with strict provenance[5]. LightRAG seamlessly ingests this data into a unified PostgreSQL or Neo4j backend, instantly providing the infrastructure for both vector similarity search and complex multi-hop graph traversal[4].
2. **ast-grep (via MCP):** This completely eliminates the need to write custom static analysis parsers for different programming languages. By utilizing the pre-built MCP server, the AI agent can query Python, TypeScript, and Rust codebases uniformly using simple YAML configuration rules[10].
3. **OpenHands:** This removes the need to build complex sandboxing environments, Docker lifecycle managers, and prompt-trimming context windows. The Agent-Computer Interface (ACI) acts as the central brain orchestrating the other tools, providing a production-ready environment for the LLM to execute its reasoning loops[52].

Production-Quality Code Worth Studying

Before writing the custom glue code required to bind these systems together, engineering teams should thoroughly analyze specific files within these repositories to understand best-in-class architectural patterns.

The **Diff Compression Algorithms** found within qodo-ai/pr-agent are vital study material. Passing an entire monolithic PR diff to an LLM immediately destroys the context window. Studying how Qodo implements chunking and summarization in its /review tool reveals how to distill a massive code change into its semantic intent[46].

The **Dual-Layer Retrieval Mechanisms** in HKUDS/LightRAG provide insight into dynamic routing. Analyzing files like llm/llama_index_impl.py demonstrates how the architecture seamlessly orchestrates an embedding model for standard semantic search alongside an LLM for graph traversal, deciding dynamically which to use based on the query complexity[4].

The **AST Pattern Unification** logic within ast-grep/ast-grep must be understood to write effective search rules. Reviewing the YAML rule engine and the implementation of relational constraints (inside, has, precedes) demonstrates how to turn a flat, syntactical tree into a highly queryable semantic structure[10].

Finally, the **Provenance Tracking** data structures in docling-project/docling represent the gold standard for explainability. Studying the ProvenanceItem schema in the DoclingDocument shows how to maintain a strict mathematical mapping back to the source text, ensuring the final Blast Radius Reports remain grounded in verifiable evidence[5].

Suggested Implementation Blueprint

The following phased implementation plan outlines the exact data flow and integration points

where the recommended open-source repositories replace custom engineering.

Phase 1: Ingestion & Knowledge Graph Foundation The architecture begins with the deployment of the underlying storage layer. A PostgreSQL database supporting pgvector and Apache AGE (or a dedicated Neo4j instance) is initialized as the persistent backend for LightRAG[36]. Crawl4AI is then deployed to autonomously traverse the organization's public application URLs, Confluence wikis, and Jira instances, generating raw, noise-free Markdown[1]. Concurrently, all internal Product Requirements Documents (PRDs), PDF specifications, and architectural diagrams are passed through the Docling pipeline. Docling extracts text, tables, and visual descriptions, ensuring absolute physical provenance for every data point[5]. This standardized data is ingested into LightRAG, which autonomously generates the foundational Product Knowledge Graph[4].

Phase 2: Codebase Static Analysis & Call Graph Indexing With the product requirements mapped, the system clones the target source code repositories. ast-grep is deployed across these repositories to perform structural indexing. Baseline YAML configurations are executed to identify and map all function declarations, API endpoints, and database queries[10]. To fuse the requirements graph with the codebase graph, Splink is utilized. It executes probabilistic matching algorithms to link the code entities found by ast-grep (e.g., checkout_api.py) to the product features documented in the LightRAG knowledge graph (e.g., "User Checkout Flow")[43]. These linkages are merged as definitive edges within the graph database.

Phase 3: The Blast Radius Agent Environment The central reasoning engine is established by deploying the OpenHands Agent-Computer Interface (ACI) within a secure Docker environment[52]. The agent is provisioned with tools: it is granted connection to the ast-grep MCP Server (allowing it to dynamically search the codebase) and API access to the LightRAG database[4]. To handle the ingestion of code changes, pr-agent is integrated via a GitHub Webhook. When a developer opens a pull request, pr-agent automatically intercepts the payload, compresses the massive git diff, and invokes the OpenHands agent[47].

Phase 4: Impact Analysis & Explainable Reporting The execution loop commences when the OpenHands agent receives the compressed diff. It utilizes its connection to the ast-grep MCP server to structurally search the codebase, determining exactly where the modified functions are called globally to establish the technical blast radius[25]. Next, the agent executes graph traversal queries against LightRAG to deduce which PRDs, User Stories, or downstream UI flows depend on the impacted technical components[4]. Finally, the agent synthesizes these findings into an explainable Blast Radius Report. Because the system maintained strict metadata lineage via Docling and Splink, the report explicitly cites the original provenance metadata to prove *why* a specific UI element or business requirement is affected. This highly accurate, evidence-backed report is then posted back to the GitHub PR via the pr-agent integration, completing the automated review cycle[24].

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